



School of Computer Science Engineering and Information Systems
FALL SEMESTER – 2026-27

Course Code: BCSE308P Course Name: - Computer Networks Lab

ASSESSMENT –3 Max Marks: 10 MARKS

General Instruction(s):

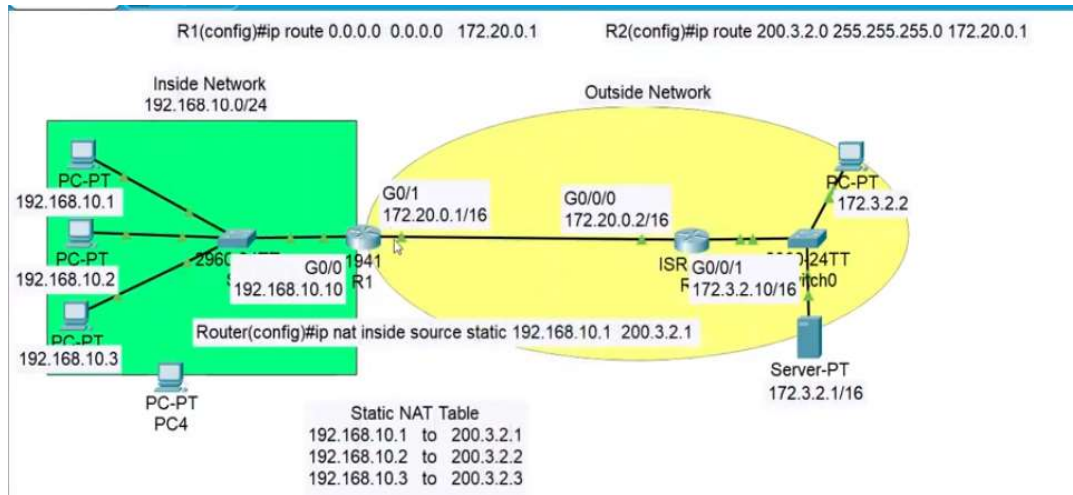
- **Student name and reg number should be written in the assignment.**
- **The students are requested to follow the below questions and submit their answers.**
 - **1. Handwritten format 2. Work with the Packet Tracer, take screen**
 - **shots, and create a PDF**
- **Final One document in PDF format should be uploaded by the students.**
- **The student should submit the assignment on or before the due date.**

1. Build a small network topology with a router using following Network configuration in Packet Tracer.

Device	Interface	IP address	Subnet mask	Gateway
Router	Fa0/0	192.168.10.1	255.255.255.0	-----
	Fa0/1	192.168.20.1	255.255.255.0	-----
PC1	NIC	192.168.10.2	255.255.255.0	192.168.10.1
PC2	NIC	192.168.20.2	255.255.255.0	192.168.20.1

2. Configure and implement a network using Packet Tracer.
 - a. Establishing a Web Server Connection Using the PC's Web Browser
 - i. What role does DNS (Domain Name System) play in establishing a connection to a web server, and how is it configured in Packet Tracer?
 - ii. What are some common issues that can prevent the PC's web browser from establishing a successful connection to the web server, and how would you troubleshoot them?

3. Implement in Packet Tracer a DHCP server with two LANs to dynamically assign IP addresses to the PC's in two LANs using an intermediate router to connect to the LANs.
4. Implement in Packet Tracer a file server(FTP) and share a file in a PC system and access the files of that folder from another PC system using an IP address
5. Create a network environment using Packet Tracer and Network Address Translation(NAT) for the following and send an email.



6. A company is granted the site address 201.70.64.0. The company needs six subnets. Design the subnets using Cisco Packet Tracer and test the connectivity using the ping command. (Intermediate device – Router)
7. A company is granted the site address 204.80.62.0 (class C). The company needs five subnets. Design the subnets using Cisco Packet Tracer and test the connectivity using the ping command. (Intermediate device – Router)